

Operating Twin for the Electric Van Transition

Where the EV ramp-up breaks the operating model, what reliability it leaks, and which controls recover the exit gate.

Scope	Single van assembly plant during a 12-month EV ramp-up
Period	January 2025 to December 2025, 53 operating weeks
Records	58,697 vehicle-orders across sequencing, yard, charging and dispatch
Method	Reproducible pipeline: DuckDB marts, Python diagnostics, scenario twin, Monte Carlo scoring
Data	Synthetic, deterministic seed. Patterns are internally consistent; absolute numbers are not measurements from a real plant.

Figures are generated directly from the processed marts and match the live operating dashboard.

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SECTION 1

Executive summary

58.7k	39%	43%	73%
VEHICLE-ORDERS ANALYSED	EV SHARE OF FLOW	DISPATCHED LATE	READY AT DISPATCH

Management should not accelerate the next EV wave through the current exit model. The plant is building to plan, but it is not shipping cleanly: the constraint has moved from production volume to readiness, staging and outbound discipline.

Executive answer	Decision logic
Decision	Hold further EV acceleration until the exit gate is controlled; implement the combined corrective package before the next ramp step.
Why now	The plant completed 58,697 orders to plan, but only 24,300 left both ready and on time, leaving 34,397 non-clean exits.
What fixes it	Cap pre-dispatch staging, reserve EV charging at shift peaks and enforce a readiness window before vehicles enter staging.
Expected effect	Scenario twin: +4.1 vehicles/day, -3.3 pp late vehicles and +1.9 decision-score points versus the unmanaged base.
Governance condition	Treat scenario impacts as pilot hypotheses until real plant data calibrates charging, staging and dispatch elasticities.

Across the full ramp-up the plant completed every scheduled order, holding a steady 160 vehicles per day while the electric mix climbed from 5 percent to nearly 80 percent of weekly output. That clears the wrong suspect. Throughput is not binding. Reliability is: 73% of completed vehicles are ready at their dispatch window, 43% of ready vehicles leave more than two hours late, and the clean-exit rate is only 41%.

The failure mode is narrow enough to act on. Electric versions are ready 41% of the time versus 93% for combustion versions, and the four EV versions carry 78 percent of all late-dispatch minutes. The physical choke point is also narrow: pre-dispatch staging runs at 43 hours p95 dwell and near-total blocking while the other yard zones clear materially faster. Internal cycle time is not the issue; EV and ICE lead-time profiles both sit around 24 hours end to end. The penalty appears after build completion, at the point where charge, state-of-charge confirmation and dispatch slots have to line up.

The priority ranking is stable under challenge. Across 300 Monte Carlo redraws of the scoring weights, Logistics ranks first 77% of the time and Yard takes the remaining 23%; no other area ever leads. The scenario twin reaches the same management answer from a different route: no single lever fixes the exit gate. The combined package, sequencing discipline plus charging capacity plus yard-buffer redesign, is the only scenario that improves throughput, internal time and late exits.

What to do, in order

1. Treat the pre-dispatch yard zone as a capacity-limited resource. Cap occupancy, redesign the buffer by destination window, and pull-sequence vehicles into staging only when their dispatch slot is real. This attacks a zone with 25.8 hours of mean dwell and 98% blocking, the physical expression of the exit delay.
2. Reserve charging slots for EV versions and add capacity at the shift peaks. Charging capacity is the highest-return lever in the scenario model at an expected impact of 0.37, and EV readiness is the gate that the corrective package moves most.
3. Impose a readiness window on dispatch. Stop releasing combustion vehicles into a staging area that is already saturated by EVs waiting on charge and state-of-charge confirmation, and the on-time gate recovers without any new throughput.

The sections that follow show the evidence chain, quantify the operating prize and separate immediate rule changes from assumptions that require calibration before capital approval.

SECTION 2

Context and objectives

A van plant in the middle of an electric ramp-up is running two factories at once. The combustion line it has optimised for years, and a growing electric line that shares the same yard, the same dispatch bays and the same shifts but behaves differently at every step. This report asks where that collision hurts and what it costs.

An EV is not a combustion vehicle with a different engine from an operations point of view. It needs a charge before it leaves, which adds a dependency that did not exist before. It carries a state-of-charge target that has to be confirmed at dispatch. It competes for a finite set of charging slots that the plant did not need a year ago. Each of these touches the back half of the flow, the part between end of line and the gate, where this plant is losing reliability.

The operating twin built for this analysis integrates five operational domains that are usually looked at separately: production sequencing, yard movements, charging sessions, dispatch readiness and outbound logistics. Looking at them together is the point. A late vehicle is rarely late for one reason. It is late because a charging slot was busy, which kept it in a staging zone, which was already full, which blocked the next vehicle. Single-domain dashboards miss that chain. The twin is designed to follow it.

Objectives

The analysis has four concrete objectives, and every section maps back to one of them.

Locate the bottleneck. Identify which area and which physical zone constrain reliable output, with evidence rather than intuition.

Quantify the EV penalty. Separate what the electric mix costs from what is baseline plant's existing operating noise.

Rank the response. Order the areas and the capacity levers by expected return, and test whether that order is stable or an artefact of the chosen weights.

Compare the futures. Put the unmanaged ramp-up side by side with targeted interventions and a combined package, on a consistent multi-criteria score.

How to read this report

Findings are organised by question, not by data source. Each finding opens with the claim, shows the chart that supports it, then explains what the chart does and does not prove. Scores reported on a 0 to 100 scale are relative pressure indices, not physical units, and are defined in the methodology section. Where a number is a modelling assumption rather than a measurement, the text says so. The recommendations in the final section are mapped to the specific findings that justify them.

Decision frame

The report is built around one management decision: whether the next EV ramp-up wave should be pushed through the current operating model, slowed until exit reliability recovers, or supported by

targeted capacity and sequencing changes. The answer is not based on one metric. It requires a chain of evidence: the plant must be shown to build volume, the exit gate must be shown to fail, the EV mix must be shown to explain a material share of that failure, and the recommended levers must improve the failure mode without creating a larger trade-off elsewhere.

Decision test	Evidence used	What would change the answer
Is throughput the binding constraint?	Daily completions, throughput gap, scenario throughput	A material negative throughput gap or falling run-rate would shift the priority back to production flow.
Is the EV mix creating a distinct penalty?	Readiness by propulsion and version; EV vs ICE pressure scores	If real plant data showed similar readiness for EV and ICE, the remedy would move from EV-specific gates to generic outbound discipline.
Is the bottleneck localised enough to act?	Yard zone dwell, OPI ranking, risk matrix and driver correlations	If congestion were evenly spread, a single staging-zone intervention would be too narrow.
Is the recommended package stable?	Scenario comparison, base-vs-corrective deltas, Monte Carlo and sensitivity tests	If Logistics and Yard stopped leading under reasonable weights, the priority order would need rework before execution.

SECTION 3

Data and methodology

Every figure in this report is reproducible from a single deterministic pipeline. The same commands that regenerate the raw tables also rebuild the marts, the diagnostics and the dashboard, so the numbers here, the numbers on the dashboard and the numbers in the source data cannot drift apart.

Data foundation

The analysis rests on 14 synthetic source tables covering orders, vehicles, versions, yard movements, charging sessions, charging slots, shifts, operational resources, bottleneck events and outbound logistics. The generator models three phases of the ramp-up, a pre-series period, the ramp itself and a stabilisation tail, so the EV share rises over time the way it would in a real launch rather than sitting at a flat average. Across the period the plant handles 58,697 vehicle-orders, of which 39% are electric, spread across three model families and eight versions, shipping to seven European markets.

The data is synthetic and seeded. That is a deliberate choice for a public, shareable analysis: it removes confidentiality constraints and makes the entire chain auditable. It also sets the boundary on interpretation. The patterns here are realistic and internally consistent, but the absolute numbers are not measurements from a real plant and should not be read as benchmarks for one.

Pipeline

Raw tables are loaded into DuckDB, which builds a staging layer, integrated views across domains, the analytical marts and a set of governed KPIs. Python then takes over for the work that SQL is poor at: feature engineering, the area-level diagnostic, the scenario twin and the scoring framework. A release gate runs last and checks structural integrity, metric consistency and the data contracts the dashboard depends on. The gate is not cosmetic. A single real dispatch with no readiness record blocks publication, because that would mean a vehicle left the plant that the data says was never ready, and the central finding of this report depends on the readiness signal being trustworthy.

Data quality

15 integrity checks run on every build, covering record cardinality, referential integrity, invalid denominators, duplicate orders and vehicles, incoherent sequencing, impossible charging sessions, out-of-range state of charge and out-of-order timestamps. All 15 pass with zero failing rows. The findings that follow sit on data that is internally clean.

Integrity check	Failing rows	Status
Cardinalidad flujo vehiculo	0	PASS
Chaves criticas nulas	0	PASS
Denominadores invalidos	0	PASS
Dispatch duplicado vehiculo	0	PASS
Duplicados ordenes	0	PASS
Duplicados vehiculos	0	PASS
Ev requiere carga sin sesion	0	PASS
Restriccion capacidad inconsistente	0	PASS
Retraso sin causa	0	PASS
Salida sin readiness	0	PASS
Scores riesgo nulos	0	PASS
Secuencia incoherente	0	PASS
Sesion carga imposible	0	PASS
Soc fuera rango	0	PASS
Timestamps fuera orden	0	PASS

Evidence map

The evidence base is layered by purpose. SQL marts carry the governed metrics; feature tables expose the operating mechanisms; diagnostic outputs turn those mechanisms into area-level priorities; scenario outputs test management choices. The report does not ask the reader to trust a black box. Each major claim can be traced to the layer designed for that job.

Analytical layer	Primary role in the report	Reader-facing use
Governed KPI mart	Single source of truth for total orders, EV share, readiness, late dispatch and dwell.	Executive summary, funnel interpretation and appendix KPI table.
Readiness mart	Version, shift and propulsion cuts for dispatch readiness and delay exposure.	EV penalty, version concentration and shift-exclusion tests.
Yard and charging features	Physical mechanism behind dwell, blocking, queueing and charging-slot pressure.	Bottleneck location, staging recommendation and charging-lever logic.
Scenario twin	Comparable operating futures under unmanaged ramp-up, single levers and combined package.	Recommended response, trade-offs, base-vs-corrective deltas.
Scoring governance	Dispersion, diversity, tier separation and ranking stability checks.	Confidence in prioritising Logistics and Yard first.

SECTION 4

Analytical framework

Three constructs carry the analysis: a set of operational pressure scores, a single Operational Priority Index that ranks the areas, and a scenario decision score. Each is defined here so that the findings can be read without reverse-engineering the maths.

Pressure scores

Raw operational metrics live on incompatible scales. A dwell time is in minutes, an occupancy rate is a fraction, a count of non-productive moves is an integer. To compare areas and propulsion types on one axis, each driver is normalised to a 0 to 100 pressure score, where higher means more operational strain. These scores are explicitly relative. A yard congestion score of 74 does not mean the yard is 74 percent full. It means the yard sits high on the observed distribution of congestion pressure. The scores answer where strain concentrates, not what the strain measures in physical units.

Operational Priority Index

The OPI combines five pressure scores into one ranking number per area: throughput loss, yard risk, charging risk, dispatch risk and launch-transition risk. It is the model's answer to a blunt management question: if we can only stabilise one or two areas next week, which ones. The index is not a causal estimate. It is a transparent, weighted prioritisation that can be recomputed under different weights; the ranking-stability test does that explicitly.

Decision score

Scenarios are scored on a multi-criteria function that rewards throughput, readiness and operational stability and penalises late dispatch and congestion risk. The score avoids single-metric optimisation, because the central finding of this report is that the plant can hit its volume target while failing its reliability target. A score that only looked at throughput would rate the unmanaged ramp-up as a success.

Governance

The prioritisation is held to five governance checks before it is allowed to drive any recommendation: the priority index must show real dispersion across areas, the risk drivers must be diverse rather than collapsing to one, the priority tiers must separate, and the top-ranked area must be stable under resampling. All five pass. The standout is dispersion: the OPI standard deviation across areas is 17.8 against a floor of 1.0, which means the ranking separates areas rather than rating them all alike.

Governance check	Value	Threshold	Status
Opi diversity	6.00	3.00	PASS
Risk driver diversity	4.00	2.00	PASS
Tier diversity	3.00	2.00	PASS
Opi dispersion	17.81	1.00	PASS
Rank stability top1 share	0.77	0.45	PASS

Interpretation rules

Three rules prevent over-reading the model. First, the OPI decides sequencing of managerial attention, not budget allocation. A score of 67 versus 65 means Logistics and Yard belong in the same intervention wave, not that one deserves exactly 3 percent more funding. Second, scenario scores rank operating configurations under a fixed set of assumptions; they do not forecast financial value. Third, any recommendation that depends on synthetic elasticities is framed as a staged operational test before it becomes a capital case.

Model output	Safe interpretation	Unsafe interpretation
Operational Priority Index	Which area should be stabilised first under the observed risk profile.	Exact monetary value, root cause proof or staff productivity rating.
Scenario decision score	Relative operating attractiveness across consistent futures.	Guaranteed improvement percentage or business-case NPV.
Pressure scores	Comparable strain across domains that use different units.	Physical utilisation percentage unless the metric is explicitly a utilisation.
Monte Carlo stability	Whether the top priority survives reasonable weight uncertainty.	Proof that the scenario elasticities are causal.

SECTION 5

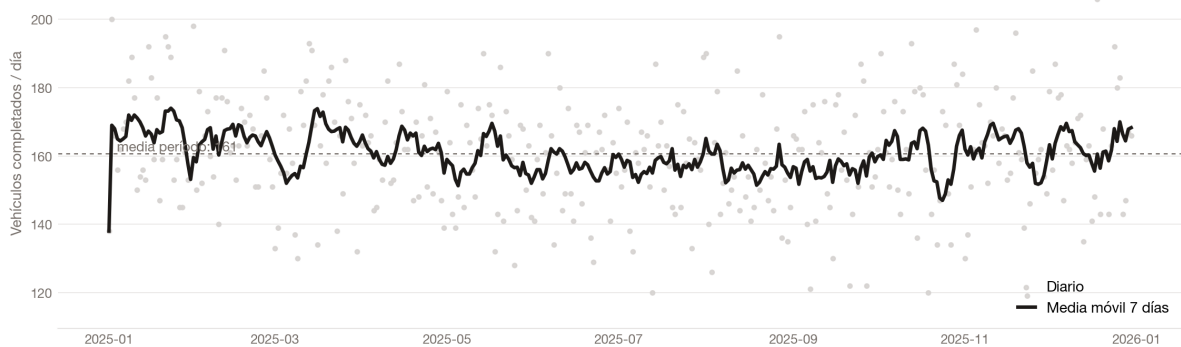
Findings: the plant builds volume but cannot ship it

Start with throughput, because it clears the wrong suspect. The plant is not throughput-constrained. It builds what it is asked to build through the entire ramp.

01 · THROUGHPUT OPERATIVO

El sistema sostiene 160 vehículos/día durante el ramp-up EV

Salida diaria del mart oficial; las olas semanales son ruido operativo, la media período es estable.



Fuente: vw_vehicle_flow_timeline · período 2025-01 a 2025-10

Gemelo Operativo EV · Synthetic factory data

Figure 1. Daily completions with a 7-day moving average. Output holds around 160 vehicles per day for the full period; the weekly waves are operating rhythm, not decline.

Daily completions sit on a flat trend at roughly 160 vehicles per day. The visible week-to-week swing is the normal breathing of a three-shift operation, not a downward drift. The throughput gap against plan is zero: every scheduled order was completed. If the only question were whether the plant can make the vehicles, the answer is yes, and the EV ramp did not change that answer.

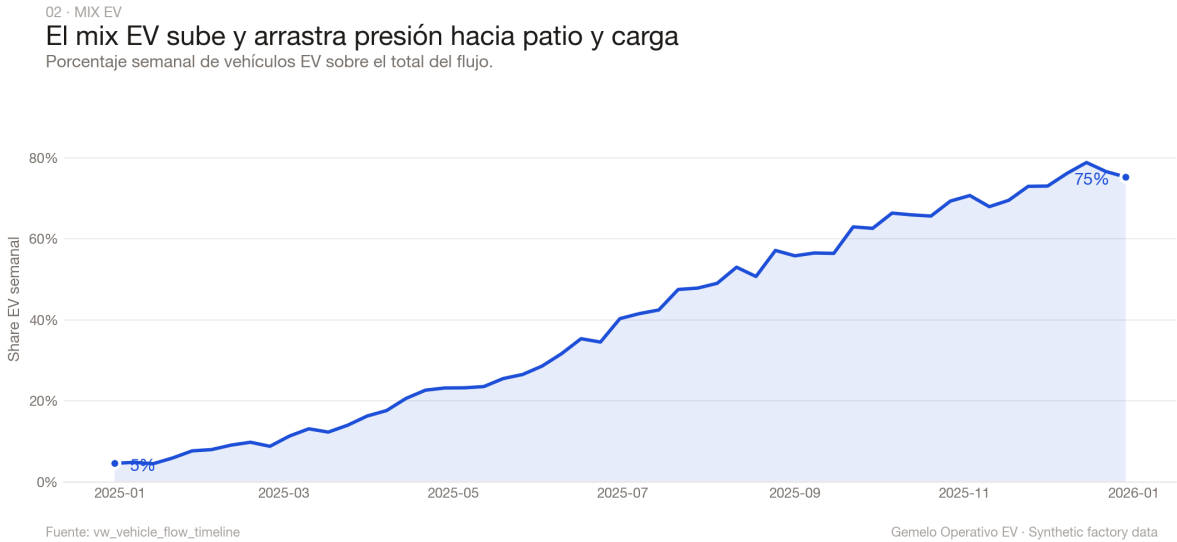


Figure 2. Weekly EV share of completed flow. The mix climbs from about 5 percent to nearly 80 percent across the ramp, a genuine transition rather than a token pilot.

The second chart shows this was a real transition, not a pilot. The electric share of weekly output rises from around 5 percent at the start to nearly 80 percent by the end. The plant absorbed that shift in mix without losing build volume. That separates execution strength from exit failure. The problem is not making EVs; it is getting them out cleanly.

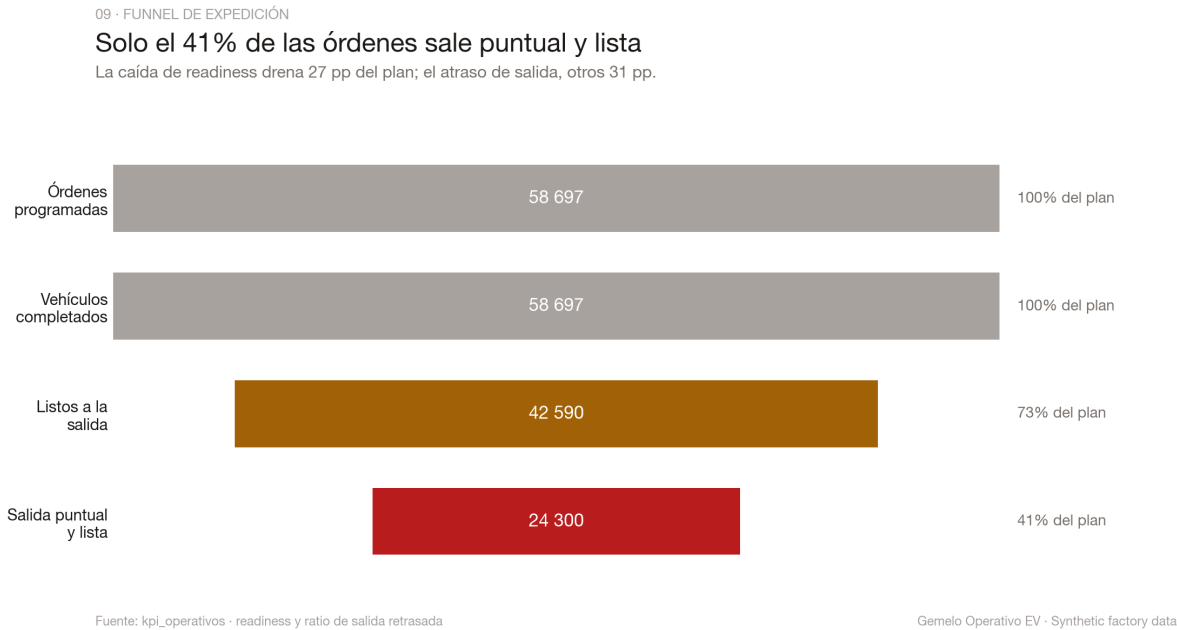


Figure 3. From scheduled order to clean exit. Readiness removes 27 points of the plan and late dispatch removes another 31, leaving a 41 percent clean-exit rate.

The exit funnel is where the volume story ends and the reliability story begins. All 58,697 orders are completed, but only 42,590 are ready at their dispatch window, a 27% readiness loss. Of those that are ready, 43% still leave late. Stack the two gates and the clean-exit rate, on time and ready, falls to 41%. The plant is converting a fully-built order book into a minority of clean exits. Every section that follows explains that collapse and sizes what recovering it is worth.

SECTION 6

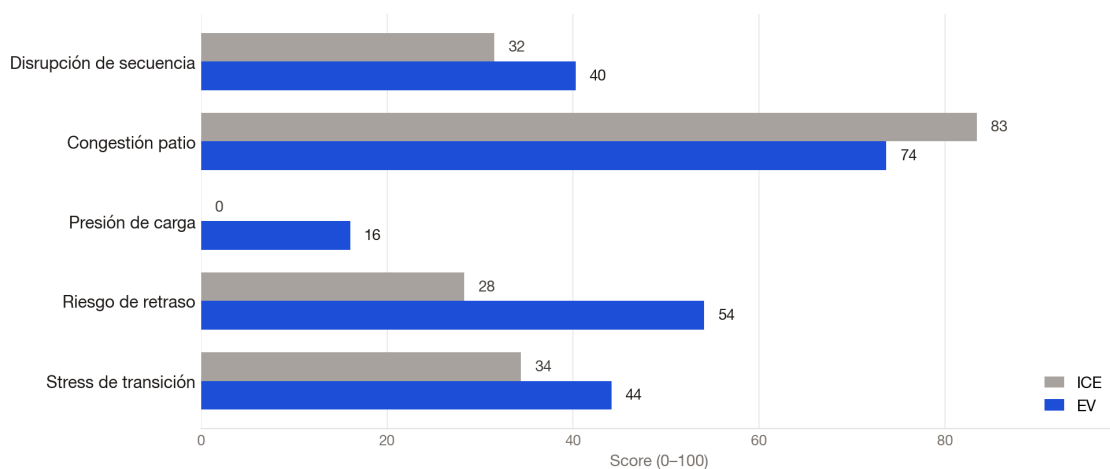
Findings: the EV penalty is real, but specific

The tempting diagnosis is to blame the ramp-up as a whole. The data is narrower. The electric mix carries a clear operational penalty, but it lands in three named places, and one obvious suspect is cleared entirely.

06 - EV VS ICE

El mix EV añade carga en secuencia, expedición y energía

Comparación por driver de presión operativa. La congestión de patio es estructural, no propia de la propulsión.



Fuente: diagnostic_ev_vs_non_ev

Gemelo Operativo EV · Synthetic factory data

Figure 4. Pressure scores by propulsion type. EV leads on sequence disruption, charging pressure and dispatch-delay risk. Yard congestion is the exception: it is structural and slightly higher for ICE.

The comparison by driver is unambiguous on three fronts. Electric vehicles score higher on sequence disruption, on charging pressure, which combustion vehicles do not generate at all, and most sharply on dispatch-delay risk, where EV sits near 54 against 28 for combustion. These are the measured transition costs, and they cluster in the back half of the flow, around charge and exit.

The exception matters as much as the rule. Yard congestion is not an EV problem. The congestion score is marginally higher for combustion vehicles, because the yard is a shared physical constraint that every vehicle passes through regardless of drivetrain. The yard is congested, but blaming the EV mix for that congestion would point the response at the wrong lever.

07 · COHORTE DE READINESS

Las versiones EV salen listas el 28–59% de las veces; las ICE, por encima del 87%

Readiness rate ponderado por volumen y versión. Ninguna versión EV se acerca al objetivo del 95%.

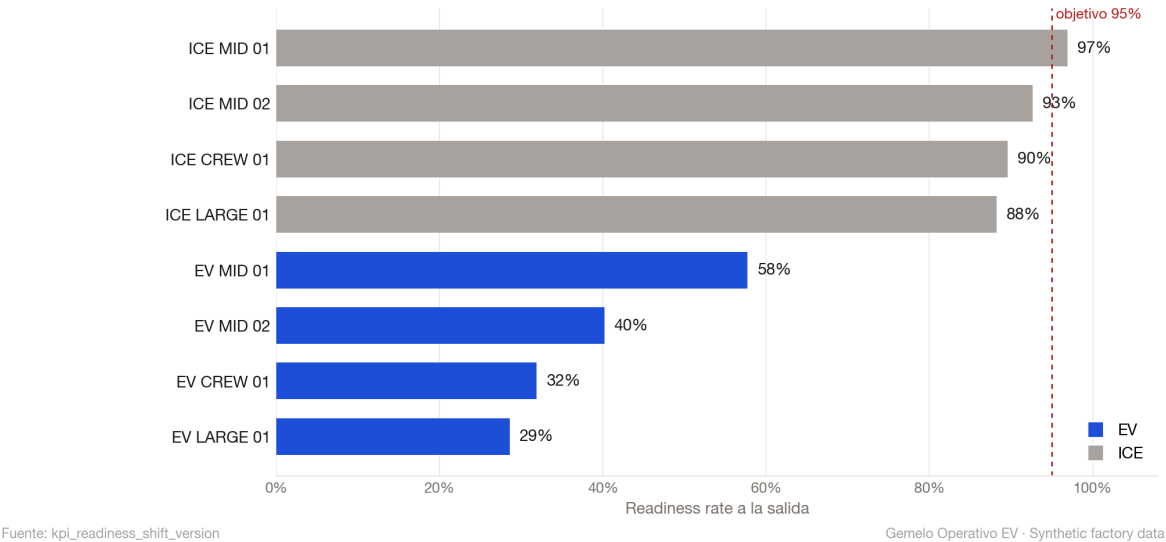


Figure 5. Readiness rate by version, weighted by volume. Every EV version sits far below the 95 percent target; every ICE version sits comfortably above 87 percent.

Readiness by version provides the cleanest split in the report. Sorted by version, the electric and combustion vehicles separate into two bands with almost no overlap. EV versions are ready between 29% and 58% of the time. Combustion versions sit between 88% and 97%. The worst EV version is VAN EV LARGE 01, at 29%; the best ICE version is VAN ICE MID 01, at 97%. No EV version comes close to the 95 percent target line. This is not a tail problem to be managed at the margin. It is a structural readiness deficit on the entire electric range, and because the EV share is rising toward 80 percent of output, it is a deficit that grows with every week of the transition.

11 · CONCENTRACIÓN DEL RETRASO

Las cuatro versiones EV acumulan el 78% del retraso total de salida

Pareto por versión. La curva acumulada confirma que el retraso no está repartido: es un problema EV.

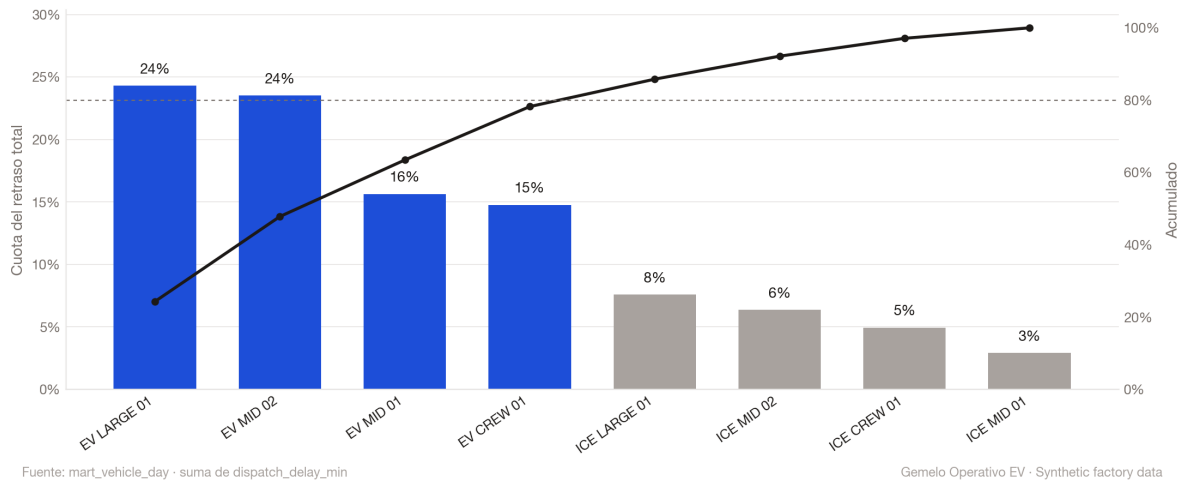


Figure 6. Concentration of total late-dispatch minutes by version. The four EV versions account for 78 percent of all delay; the cumulative curve confirms the problem is not spread across the range.

If readiness is the mechanism, late-dispatch minutes are the consequence, and they concentrate where the readiness gap predicts. The four electric versions carry 78 percent of all late-dispatch minutes. The Pareto curve rises steeply and then flattens, which is the signature of a concentrated problem rather than a diffuse one. The plant does not need to improve everything at once; it needs to move the readiness of four versions.

10 · DISTRIBUCIÓN DE LEAD TIME

EV e ICE comparten perfil de lead time interno (24 h vs 25 h)

Tiempo interno de extremo a extremo. El coste EV no está en el tiempo de ciclo, sino en readiness y salida.

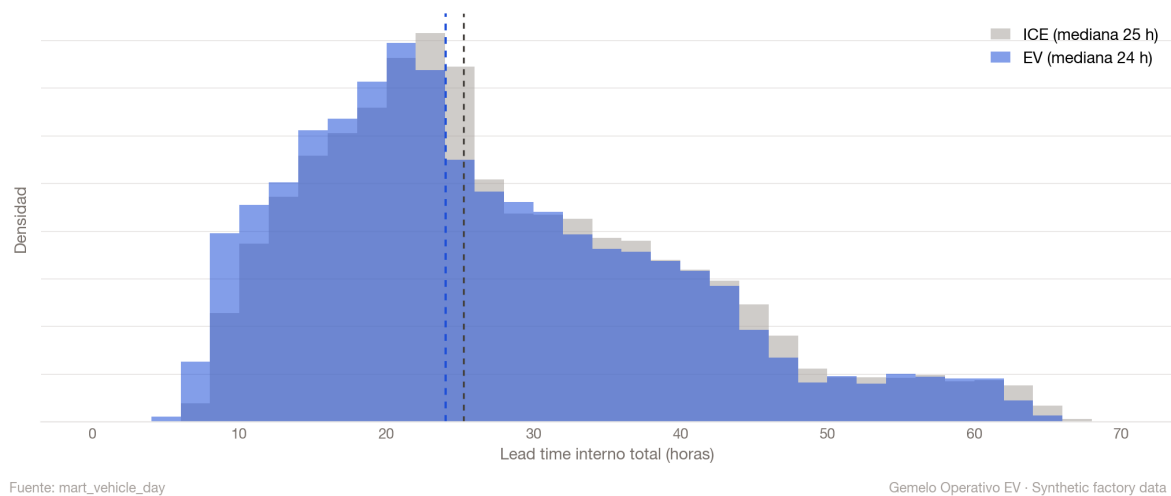


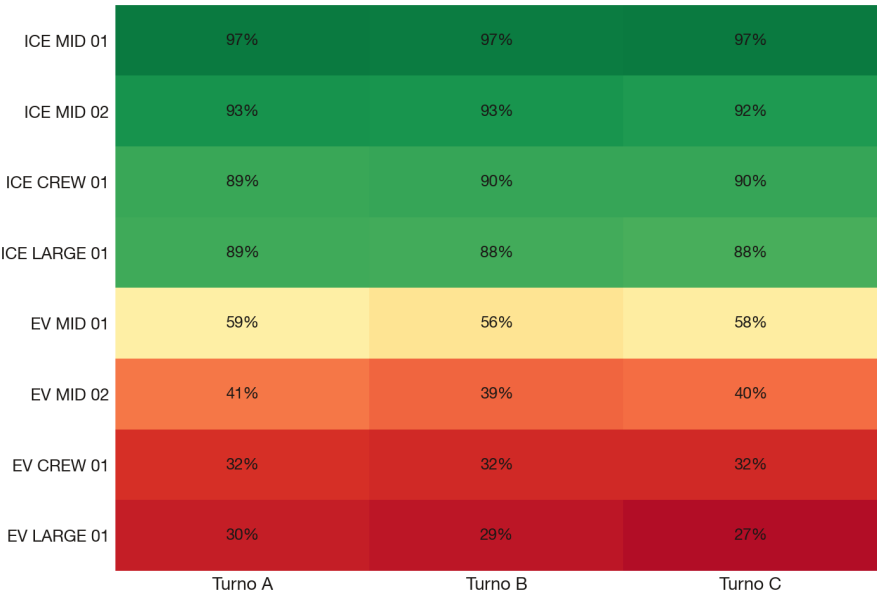
Figure 7. Internal lead-time distributions overlap almost completely, with medians of 24 and 25 hours. The EV penalty is not in cycle time.

The cleared suspect is cycle time. It would be reasonable to assume electric vehicles take longer to move through the plant. They do not. The internal lead-time distributions for EV and ICE sit almost on top of each other, with medians of 24 and 25 hours and near-identical 90th percentiles. The electric vehicles are built and moved at the same pace as the combustion ones. What happens to them is different: they reach the end of the line on time and then fail to leave, because the charge, the state-of-charge confirmation and the staging space are not there when they need them. This is why the report keeps separating build performance from exit performance. They are different problems with different owners.

18 · READINESS POR TURNO × VERSIÓN

El hueco de readiness es de versión, no de turno: los tres turnos se comportan igual

Readiness rate por versión y turno. La variación entre turnos es marginal frente al salto EV→ICE.



Fuente: kpi_readiness_shift_version

Gemelo Operativo EV · Synthetic factory data

Figure 8. Readiness by version and shift. The colour bands run horizontally, not vertically: readiness is a property of the version, not the shift that built it.

One more alternative explanation deserves a direct test: maybe a particular shift is the weak link. The heatmap rules it out. Colour varies down the chart, by version, and barely varies across it, by shift. Shifts A, B and C produce nearly identical readiness for any given version. The deficit travels with the product, not the crew, which means the fix is a process and capacity fix, not a training or staffing fix aimed at one shift.

What the EV penalty is, and what it is not

The weighted readiness gap between EV and ICE is 51.6 percentage points across 22,997 electric and 35,700 combustion orders. That is large enough to dominate the dispatch funnel, but the surrounding diagnostics narrow its meaning. It is not a production-cycle penalty, because lead-time distributions overlap. It is not a shift penalty, because A, B and C show the same version pattern. It is not a market allocation penalty, because destination readiness is broadly flat. It is a readiness-at-exit penalty caused by charge dependency, state-of-charge confirmation and the physical staging buffer absorbing vehicles that are not yet dispatchable.

Hypothesis tested	Result	Decision implication
EVs take longer to build or move internally	Rejected: EV and ICE lead-time distributions overlap around the same 24-25 hour centre.	Do not attack this first through production cycle-time projects.
A weak shift is driving poor readiness	Rejected: readiness varies by version far more than by shift.	Do not localise the fix to one crew; standardise the exit process across all shifts.
One destination market is creating the delay	Rejected: readiness is materially uniform across markets.	Do not solve through market allocation before fixing internal dispatch readiness.
EV readiness is structurally below ICE readiness	Confirmed: EV readiness is 41% against 93% for ICE.	Prioritise charging reservation, SOC confirmation and EV-specific dispatch gates.

SECTION 7

Findings: where the bottleneck physically is

The EV penalty explains who is late. It does not by itself explain where vehicles get stuck. For that the analysis needs the physical yard view and the area-level diagnostic.

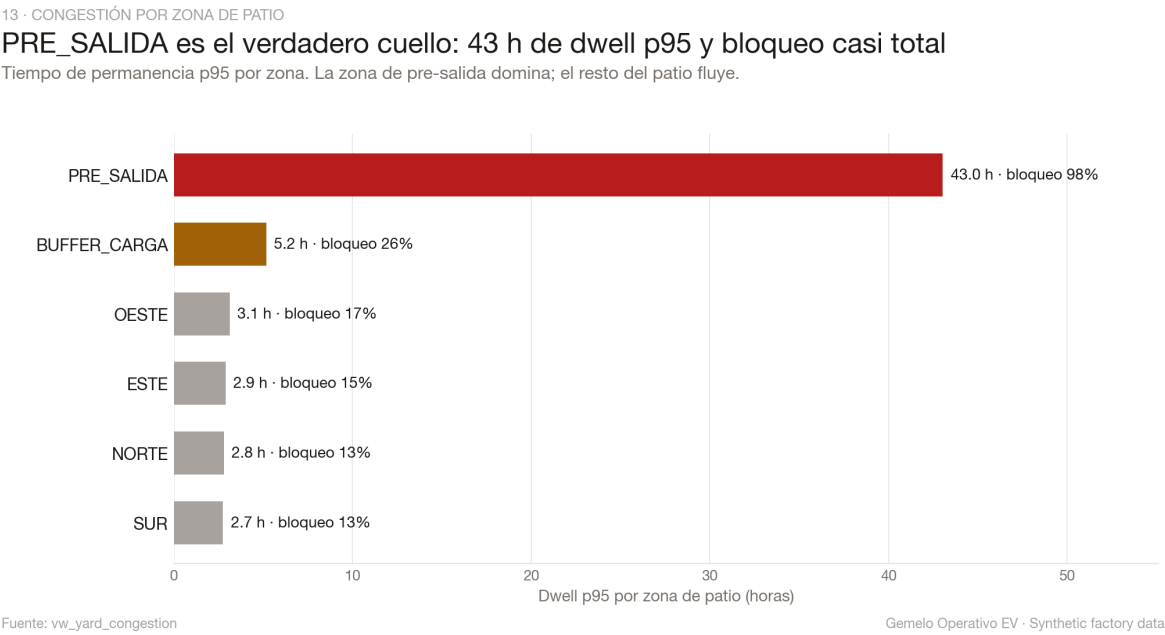


Figure 9. p95 dwell time by yard zone. Pre-dispatch staging dominates at roughly 43 hours and near-total blocking; the next-highest zone, buffer charging, clears in about five hours, and the remaining four zones all clear in under three-and-a-half.

The yard is not uniformly congested. One zone carries the entire problem. The pre-dispatch staging area shows a mean dwell of 25.8 hours, a p95 observation above 43.0 hours and a blocking rate of 98%, while the next-highest zone, Buffer Carga, averages 3.5 hours. Vehicles that are not yet ready accumulate in staging, the staging area saturates, and the blockage delays vehicles behind them, including ready ones. The plant-level p95 dwell of 49 hours is driven by this one zone.

03 · PRIORIZACIÓN OPERATIVA

LOGÍSTICA y PATIO concentran la presión esta semana

OPI por área: combina pérdida de throughput, riesgo de salida y stress operativo.

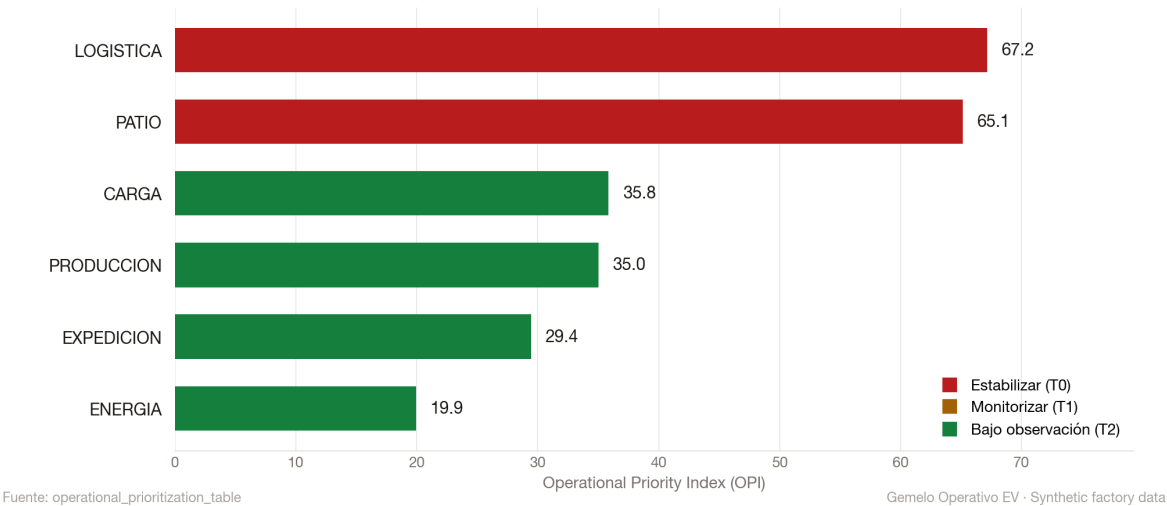


Figure 10. Operational Priority Index by area. Logistics and Yard form a clear top tier; Charging, Production, Dispatch and Energy fall well below.

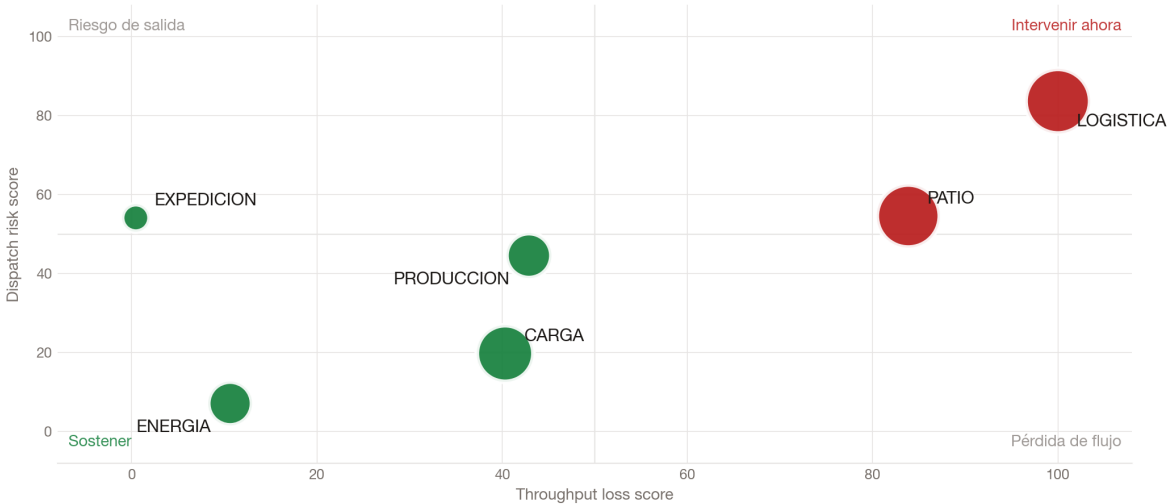
Lifted to the area level, the Operational Priority Index puts Logistics and Yard in a tier of their own, at 67 and 65, with the next area more than 25 points behind. This reconciles with the yard data. Logistics owns the dispatch-readiness and outbound-window failures; Yard owns the physical staging saturation. The two are the same failure seen from two angles, which is why they rank together and why the recommendations treat them as one coordinated intervention rather than two separate projects.

Area	OPI	Main risk driver	Recommended action	Tier
LOGISTICA	67.2	throughput loss score	Ajustar turnos o recursos	Stabilise next wave
PATIO	65.1	yard risk score	Revisar política de buffer en patio	Stabilise next wave
CARGA	35.8	charging risk score	Ampliar infraestructura de carga	Watch
PRODUCCION	35.0	dispatch risk score	Priorizar expedición selectiva	Watch
EXPEDICION	29.4	dispatch risk score	Priorizar expedición selectiva	No immediate priority
ENERGIA	19.9	charging risk score	Ampliar infraestructura de carga	No immediate priority

04 · MATRIZ DE RIESGO

Pérdida de flujo × riesgo de salida — dónde actuar primero

Tamaño del punto: operational stress. LOGÍSTICA y PATIO están en el cuadrante de intervención inmediata.



Fuente: operational_prioritization_table

Gemelo Operativo EV · Synthetic factory data

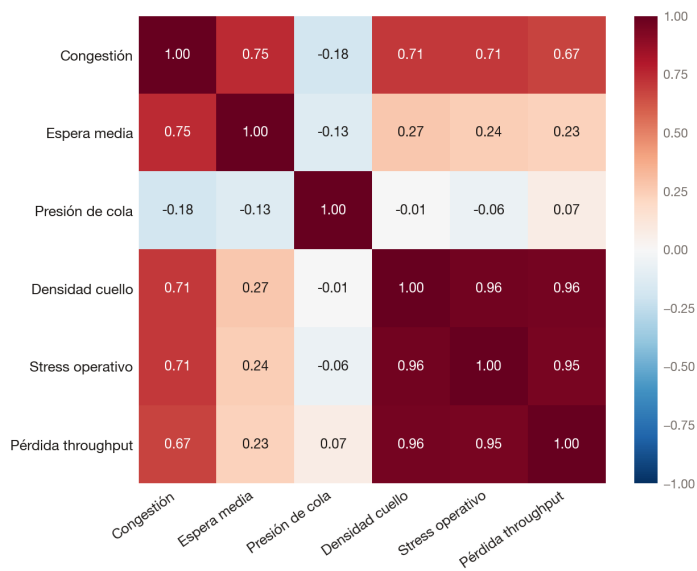
Figure 11. Throughput loss against dispatch risk by area, sized by operational stress. Logistics and Yard sit in the intervene-now quadrant; Energy and Dispatch sit in the sustain quadrant.

The risk matrix is the one-slide version of the priority story. Plotting throughput loss against dispatch risk, Logistics lands in the top-right intervene-now quadrant on throughput loss, Yard sits high on the same axis with heavy operational stress, and the remaining areas fall toward the sustain corner. Position encodes risk on both axes; bubble size encodes stress. The two areas that require executive attention are visible without reading the table.

17 · CORRELACIÓN DE DRIVERS

Congestión, espera y stress se mueven juntos; la pérdida de throughput es casi independiente

Correlación de Pearson entre drivers a nivel área-turno. Confirma dos familias de riesgo distintas.



Fuente: mart_area_shift

Gemelo Operativo EV · Synthetic factory data

Figure 12. Correlation between operational drivers at area-shift level. Congestion, waiting and stress move together; throughput loss is nearly independent of them.

The correlation structure explains why the priority list has two distinct heads rather than one. Congestion, average waiting time, queue pressure and operational stress form a tight cluster that moves together: these are the physical yard family of risk. Throughput loss sits almost orthogonal to that cluster, which means it is driven by something else, the readiness-and-dispatch family. Two independent risk families is precisely why Logistics and Yard both reach the top tier through different drivers, and why a single-lever fix cannot resolve both.

SECTION 8

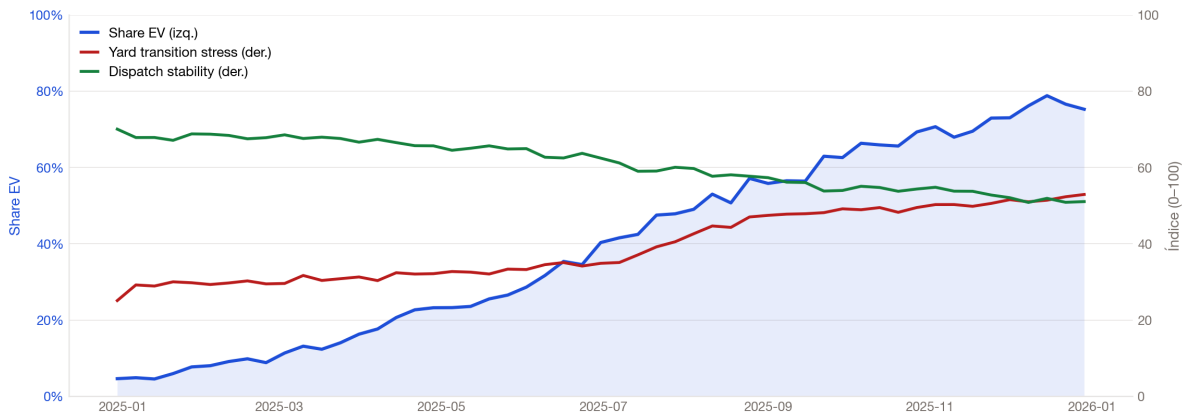
Findings: the pressure rises with the mix

A single-period snapshot can hide whether a problem is getting worse. The weekly transition series shows that it is, and that the deterioration tracks the EV share closely.

08 · TRANSICIÓN DE LANZAMIENTO

El stress de patio sube con el mix EV; la estabilidad de salida cae en paralelo

Series semanales del ramp-up (53 semanas). El stress de transición pasa de 25 a 53 cuando el EV domina el flujo.



Fuente: launch_transition_features

Gemelo Operativo EV · Synthetic factory data

Figure 13. Weekly EV share against yard-transition stress and dispatch stability. As the mix rises, yard stress climbs from 25 to 53 and dispatch stability erodes in parallel.

The weekly series adds the time dimension. As the electric share climbs by 70.7 percentage points across the 53 weeks, the yard-transition stress index rises by 27.8 points and the dispatch-stability index falls by 19.0 points. The back-of-line pressure is not a fixed cost of running EVs; it scales with how many of them are in the flow. Left unmanaged, the trend says the exit problem gets worse as the plant succeeds at its transition target, which is the most dangerous shape a problem can take because success in one metric manufactures failure in another.

The charging-capacity gap also moves in the wrong direction, widening by 14.5 points from the first to the final operating week. That does not mean chargers are fully utilised all year; the average utilisation reported in the governed KPI mart is only 14.8%. It means the constraint is temporal concentration: charging demand arrives at the wrong time relative to slot availability, and that mismatch compounds as the electric share rises. The remedy therefore has to include reservation and peak-shift capacity, not a year-average utilisation target.

12 · MERCADOS DE DESTINO

Iberia concentra un tercio del volumen; el readiness es uniforme entre destinos

Volumen por mercado y readiness final (color verde = más listo). El riesgo es operativo, no geográfico.

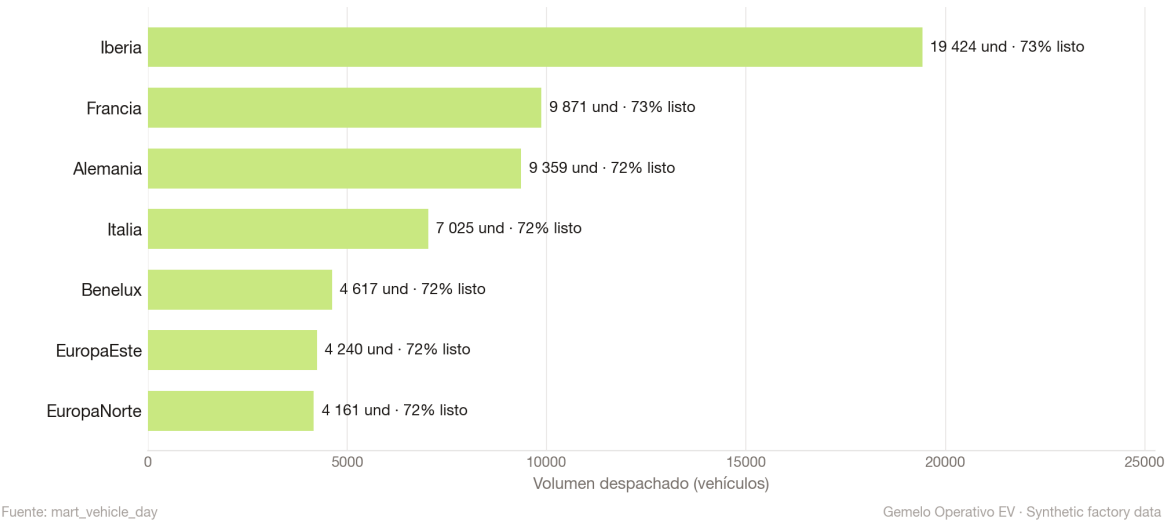


Figure 14. Volume and readiness by destination market. Iberia takes roughly a third of output; readiness is uniform across destinations.

Geography is the second cleared suspect. Output is concentrated in Iberia, which takes around a third of volume, followed by France and Germany, but readiness is essentially uniform across all seven destinations. No market is structurally worse served than another. The fix is internal and operational, not a matter of reallocating volume between destinations or renegotiating outbound windows market by market. The lever is inside the plant.

SECTION 9

Findings: which response works

The diagnostic says what is wrong and where. The scenario twin asks what to do about it, by simulating eight futures on a consistent decision score and letting them compete.

05 · COMPARADOR DE ESCENARIOS

La combinación de medidas correctivas gana al ramp-up puro

Decision score = throughput + readiness + estabilidad – riesgo de salida.

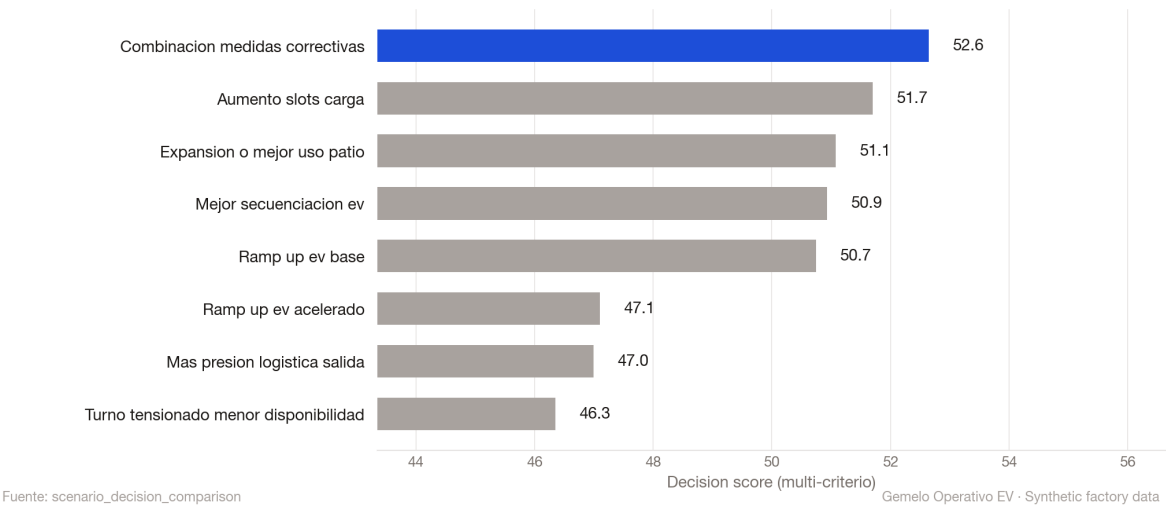


Figure 15. Decision score across eight scenarios. The combined corrective package leads; accelerating the mix without support scores worst.

The combined corrective package wins, at a decision score of 52.6. The unmanaged base ramp-up scores 50.7, and the worst option, accelerating the EV mix with no operational support, falls to 46.3. The ordering carries a clear message. Pushing more electric volume into the existing configuration is the worst move available, because it loads the exact gate that is already failing. The single-lever scenarios, more charging slots, better sequencing, more yard space, each help a little and cluster in the middle. Only the package that does all three together separates from the pack.

14 · TRADE-OFF DE ESCENARIOS

La combinación correctiva es la frontera: más throughput con menos retraso

Cada punto es un escenario. Arriba-izquierda es mejor (eje X invertido). El tamaño es el decision score.

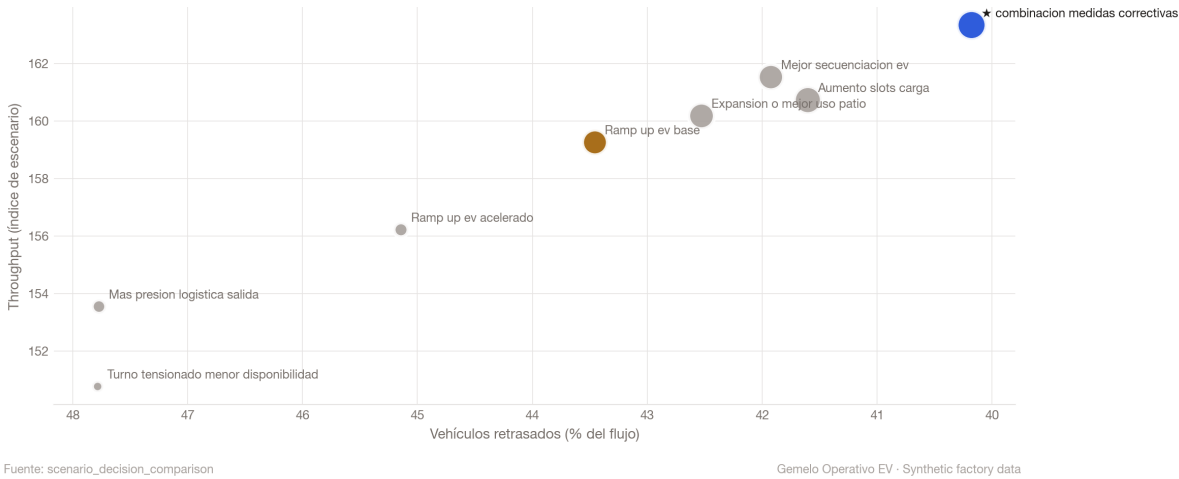


Figure 16. Throughput against late vehicles by scenario, with the x-axis inverted so better is up and to the left. The corrective package sits on the efficient frontier.

The trade-off view shows why. Plotting throughput against the share of late vehicles, the scenarios trace a frontier, and the corrective package sits at its best corner: higher throughput with fewer late vehicles at the same time. The accelerated-mix scenario sits at the opposite corner, buying nominal throughput with a sharp rise in late exits. This removes the volume-first option from the decision set. On this data, faster without support moves the plant the wrong way on the metric that is already broken.

Scenario	Throughput	Internal time	Late veh.	Stability	Decision
Combinacion medidas correctivas	163	1592	40%	88	52.6
Aumento slots carga	161	1640	42%	88	51.7
Expansion o mejor uso patio	160	1657	43%	87	51.1
Mejor secuenciacion ev	162	1644	42%	88	50.9
Ramp up ev base	159	1683	43%	87	50.7
Ramp up ev acelerado	156	1718	45%	86	47.1
Mas presion logistica salida	154	1735	48%	86	47.0
Turno tensionado menor disponibilidad	151	1758	48%	86	46.3

19 · ANTES VS DESPUÉS

El paquete correctivo mejora throughput, estabilidad y tiempo interno a la vez

Escenario base de ramp-up frente a la combinación de medidas. Verde = mejora, rojo = deterioro.

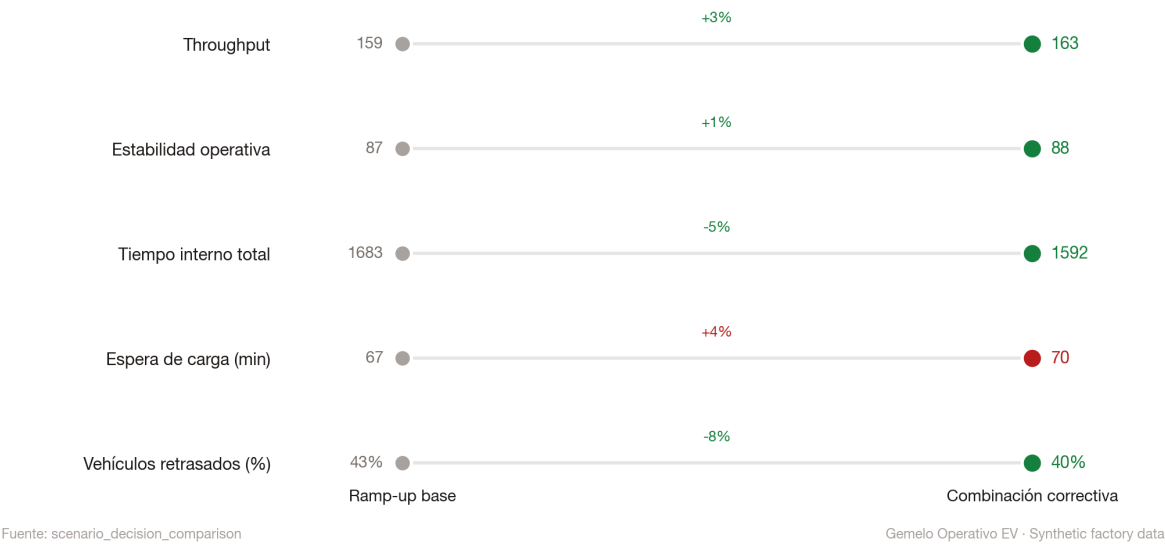


Figure 17. The base ramp-up against the corrective package across five metrics. Throughput, stability, internal time and late vehicles all improve; charging wait edges up as the only trade.

Held directly against the unmanaged base, the corrective package improves throughput by 2.6 percent, lifts operational stability, cuts internal time by 5.4 percent and reduces late vehicles by 3.3 points. The one metric that moves the wrong way is average charging wait, which rises by 4.0 percent. That trade-off is material: pulling more EVs through readiness on time puts more simultaneous demand on chargers. It is also why charging capacity is the first lever to test.

15 · RANKING DE PALANCAS

La capacidad de carga es la palanca de mayor retorno esperado

Impacto relativo de cada palanca operativa. Secuenciación y gestión de patio la siguen de cerca.

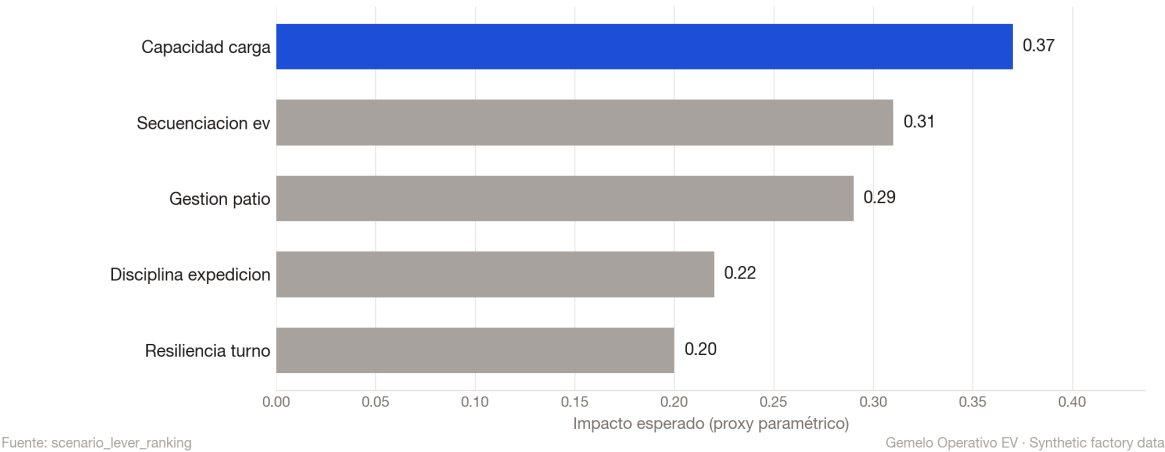


Figure 18. Capacity levers by expected impact. Charging capacity leads, with sequencing and yard management close behind.

Ranked individually, charging capacity is the highest-return lever at an expected impact of 0.37, with EV sequencing at 0.31 and yard management at 0.29. The gaps between the top three are small, which is the quantitative reason the combined package beats any single move. No one lever dominates by enough to stand alone. The impact figures are parametric assumptions inside the scenario model, not measured elasticities, and the recommendations section treats them as a priority ordering rather than as promised percentages.

Base versus corrective package

The base-to-corrective comparison is the most decision-useful scenario readout because it isolates the recommended package from the unmanaged ramp-up. The package improves the metrics that matter most to the operating thesis: more vehicles flow through the system, internal time falls, congestion risk falls and the late-vehicle rate drops. The increase in average charging wait is the only negative movement, and it is acceptable only if the package is paired with peak charging reservation rather than treated as a pure yard or sequencing project.

Metric	Base	Corrective package	Movement
Throughput	159.25	163.33	+2.6%
Internal time	1683	1592	-5.4%
Congestion risk	0.097	0.091	-6.0%
Late vehicles	43.5%	40.2%	-7.5%
Charging wait	67.1 min	69.8 min	+4.0%

SECTION 10

Findings: is the ranking trustworthy

A prioritisation that flips the moment someone questions the weights is worthless for decision-making. The scoring framework is stress-tested so that the top of the ranking can be defended.

16 - ROBUSTEZ DEL RANKING

LOGÍSTICA encabeza el ranking en el 77% de los remuestreos Monte Carlo

300 sorteos Dirichlet sobre los pesos del score. La conclusión no depende de una única ponderación.

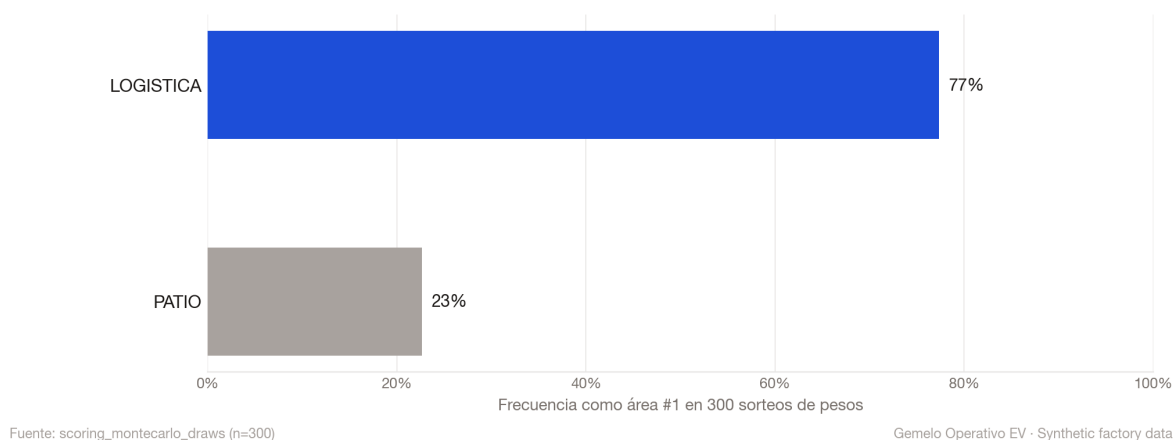


Figure 19. Top-ranked area across 300 Monte Carlo redraws of the scoring weights. Logistics leads 77 percent of the time and Yard takes the rest; no other area ever reaches first.

The weights in the priority index are a judgement call, so the analysis redraws them 300 times from a Dirichlet distribution and recomputes the ranking each time. Logistics comes first in 77 percent of the draws and Yard in the remaining 23 percent. No third area ever takes the top spot. The headline conclusion of this report, that Logistics and Yard are the two areas to stabilise, does not depend on one particular set of weights. It holds across the full range of reasonable weightings, and the only thing that moves is the order of the top two, both of which are recommended for coordinated action anyway.

The sensitivity analysis reinforces this from a different direction. Perturbing each individual risk driver up and down by 20 percent leaves Logistics and Yard in the top two of the top-three list in every single case. The mean score shifts only at the second decimal. A ranking this stable under both weight resampling and driver perturbation is one a plant manager can act on without treating the recommendation as an artefact of one modelling choice.

Stress test	Observed result	Interpretation
Monte Carlo weight redraws	Logistics top-1 in 77.3%; Yard top-1 in 22.7%; no other area first.	The recommended first wave is stable even if the exact order of Logistics and Yard is debated.
Yard-risk perturbation	Top three remains Logistics, Yard and Charging under both -20% and +20% factors.	The yard recommendation is not dependent on overstating its own risk score.
Charging-risk perturbation	Charging and Production trade the third slot; Logistics and Yard remain first two.	The first-wave recommendation is unaffected; charging remains an enabling lever rather than the top area.
Dispatch-risk perturbation	Logistics and Yard remain first two under both directions.	The dispatch-readiness diagnosis survives uncertainty in the dispatch-risk driver.

Residual uncertainty

The remaining uncertainty is not about which two areas lead. It is about the size of the benefit after intervention. That distinction matters. The diagnostic is strong enough to start operational changes immediately, because it is anchored in observed synthetic flow mechanics and stable rankings. The business case for new physical chargers or yard expansion would still require calibration on real plant data, because the scenario elasticities are assumptions. Start with sequencing, reservation and staging discipline now; use the first implementation cycle to estimate the real elasticity before committing irreversible capital.

SECTION 11

Risks, limitations and caveats

A credible report is explicit about where it stops. The following limitations bound how far these findings should be pushed, and none of them is hidden in a footnote because each one changes how the results should be used.

The data is synthetic

This is the first and largest caveat. The records are generated, not measured. The structure is realistic and the relationships are internally consistent, which is enough to demonstrate the method and to show what such an analysis would surface on real data. It is not enough to treat any absolute number here as a fact about a physical plant. The 43-hour staging dwell and the 41% clean-exit rate are properties of this dataset, not benchmarks.

Scenario impacts are assumptions, not estimates

The lever impacts and the scenario outcomes come from parametric elasticities chosen to be plausible. They are not causal estimates from an experiment or a natural intervention. The scenario results should be read as a structured way to compare options under stated assumptions, and the ranking they produce is more trustworthy than any individual percentage they report. Before any capital commitment, these elasticities would need calibration against the plant's own response to past changes.

Scores are relative, not physical

Every 0 to 100 score in this report is a position on a distribution, not a measurement in minutes, units or euros. They are built for comparison across areas and propulsion types, which they do well. They are not built to be read as absolute levels and should not be quoted as such outside the comparative context that gives them meaning.

The throughput-loss proxy is attribution, not causation

The area throughput-loss figure attributes lost flow to the bottleneck events recorded against an area. It is a proxy that apportions observed loss, not a counterfactual estimate of what output would have been without the bottleneck. It is reliable for ranking where loss concentrates and should not be over-read as a precise recoverable-output number.

Weights and thresholds need local calibration

The 95 percent readiness target, the 120-minute late-dispatch threshold and the scoring weights are defaults. They are reasonable and the conclusions hold when they move within sensible ranges, as the ranking-stability section shows, but a real deployment would set them to the plant's own standards and service-level agreements before the numbers drive money.

The dashboard needs network access

The companion dashboard is a single self-contained HTML file with all data embedded, but it loads its charting library and web fonts from a content delivery network. It renders fully online and degrades gracefully offline. For a fully air-gapped deployment those two dependencies would need to be

vendored into the file.

SECTION 12

Recommendations and action priorities

The recommendation is not a broad transformation. It is a five-part control package: three operating changes that recover the exit gate, one transition-governance decision, and one calibration step before capital approval.

#	Action	Owner	Tied to
1	Cap and redesign the pre-dispatch staging zone; pull-sequence into staging only against real dispatch slots	Yard / Logistics	Fig 9, 11
2	Reserve charging slots for EV versions and add capacity at shift peaks	Energy / Charging	Fig 4, 5, 18
3	Enforce a readiness window on dispatch; hold ICE out of a saturated staging area	Logistics	Fig 3, 5
4	Stop accelerating the EV mix until exit reliability recovers	Plant / Planning	Fig 13, 15, 16
5	Calibrate weights, thresholds and elasticities to plant standards before capital decisions	Operations analytics	Section 11

Priority 1. Fix the staging chokepoint

The pre-dispatch staging zone is the physical bottleneck, at 43 hours p95 dwell and near-total blocking against five hours or less everywhere else in the yard. Treat it as a capacity-limited resource rather than open ground. Set a hard occupancy cap, segment the buffer by destination dispatch window, and only pull a vehicle into staging when its dispatch slot is confirmed. This is the highest-confidence recommendation because it rests on measured dwell data, not on a modelling assumption, and because it directly relieves the blocking that delays even ready vehicles.

Priority 2. Unblock EV readiness through charging

EV readiness is the gate, charging is its dominant input, and charging capacity is the highest-return lever in the scenario model. Reserve slots for electric versions so they are not competing with discretionary demand, and add capacity at the shift peaks where the queue forms. Average charger utilisation across the period is only 15%, which says the constraint is timing and allocation at the peaks, not a shortage of total charging energy. The highest-pressure charging zone is Norte, with 6997 sessions and an average pressure score of 28.7. The first intervention can therefore be a slotting rule before it becomes an infrastructure programme.

Priority 3. Put a readiness gate on dispatch

A large part of the exit problem is sequencing discipline, not capacity. Combustion vehicles, which are ready more than 90 percent of the time, are being released into a staging area already saturated with EVs waiting on charge and state-of-charge confirmation. A readiness window that holds vehicles

out of staging until their slot is real decongests the zone without adding a single square metre or a single charger. It is the lowest-cost item in the package and it supports the other two.

Priority 4. Do not accelerate the mix into a broken gate

The transition trend and the scenario ranking agree that pushing more EV volume into the current configuration is the worst available move. Hold the planned mix steady until the exit gate recovers. This is a sequencing decision about the transition itself, and it costs nothing except patience, while protecting the plant from manufacturing more late exits in the name of hitting an electrification milestone.

Priority 5. Calibrate before you commit capital

Before any of the above drives a spending decision, calibrate the model to the plant. Replace the default readiness target and late threshold with the plant's own service levels, and fit the scenario elasticities to the plant's observed response to past operational changes. The diagnostic and the ranking are strong enough to act on now. The scenario magnitudes are not, and should earn their place against real history before they size an investment.

The size of the prize

To frame the stakes in one line: the plant already builds 58,697 vehicles to plan, but only about 24,300 of them leave both on time and ready today. The corrective package does not ask the plant to build more; it focuses on shipping cleanly what the plant already builds. That is high-return operational improvement, because the volume has already been paid for. Recovering even half of the clean-exit gap would do more for the plant's reliability than any throughput initiative, at a fraction of the cost.

Implementation roadmap

The recommendations should not be launched as one large programme. The first wave should change rules and operating discipline, because those actions are reversible and will generate the calibration data needed for any later capital case. Physical capacity should be the second wave, not the starting point, unless the pilot proves that the charging or staging constraint remains binding after sequencing and reservation are enforced.

Horizon	Actions	Exit criteria
0-2 weeks	Freeze acceleration of EV mix; define readiness window; introduce staging occupancy cap; publish daily clean-exit metric.	Clean-exit rate tracked daily; no vehicle pulled into pre-dispatch staging without confirmed dispatch slot.
2-6 weeks	Reserve EV charging slots by version and shift; rebalance pre-dispatch buffer by outbound window; run shift-level control room review.	EV readiness improves week over week; pre-dispatch blocking rate falls; no deterioration in total throughput.
6-12 weeks	Test peak charging capacity and staffing changes; compare observed elasticity against scenario assumptions.	Measured response supports or rejects charger expansion and yard redesign investment case.
Before capital	Recalibrate scenario model with plant history; refresh OPI weights with operations leadership; rerun release gate.	Decision pack separates confirmed operational gains from assumptions requiring investment approval.

Management controls

The control set should be small enough to run every day. Track clean-exit rate, EV readiness rate, pre-dispatch staging blocking, peak charging wait, late-dispatch minutes by version and OPI top-two stability. The first four metrics tell the plant whether the intervention is working; the last two prevent the organisation from declaring victory because one symptom improved while the root bottleneck moved somewhere else.

Control	Owner	Operating trigger
Clean-exit rate	Plant manager	Daily rate fails to improve for five operating days after staging and readiness rules go live.
EV readiness rate	Logistics / Charging	EV versions remain below the agreed target after reserved slots are introduced.
Pre-dispatch blocking	Yard	Blocking does not fall after occupancy caps and destination-window segmentation are enforced.
Peak charging wait	Energy / Charging	Wait time rises while readiness improves, signalling the point where slotting is no longer enough.
Late minutes by version	Operations analytics	Delay concentration shifts away from the four EV versions, indicating the bottleneck has moved.
OPI top-two stability	Operations analytics	Logistics and Yard stop ranking in the top two under the latest operating data.

SECTION 13

Further questions for real deployment

The report is decision-ready for a synthetic operating twin and method demonstration. A real plant deployment would need five additional answers before the analysis could support capital approval or contractual commitments.

Question	Why it matters	Evidence needed
What is the plant's true service-level definition of a clean exit?	The report uses readiness plus a two-hour late threshold; the plant may manage to different customer windows.	Outbound SLA, carrier window rules, customer penalty logic and accepted exception codes.
Which charging failures are capacity, scheduling or SOC-confirmation failures?	The operating response differs: add hardware, reserve slots or change inspection/sign-off process.	Session timestamps, charger availability, SOC target misses, interruption reasons and manual override logs.
How much staging dwell is avoidable versus policy-driven?	Some dwell may be intentional batching for outbound routes; only avoidable dwell should drive a redesign case.	Dispatch-slot assignment, destination grouping, carrier arrival records and yard movement intent codes.
What elasticity does each intervention show in practice?	Scenario impacts are assumptions until observed on the plant's own process.	Pilot design with pre/post windows, stable demand mix and measured response on readiness, dwell and late exits.
How should the scorecard weight cost, capex and service penalties?	The current model ranks operational outcomes, not financial return.	Cost of chargers, yard changes, overtime, carrier penalties, working-capital impact and customer-service penalties.

None of these questions blocks the operating recommendations. They do block a capital business case. That is the right governance boundary: use the diagnostic to fix rule-based leakage now, and use the implementation period to turn the synthetic scenario twin into a calibrated investment model.

SECTION 14

Appendix

A. Operational KPI summary

Metric	Value
Total vehicle-orders	58,697
Throughput gap to plan	0
EV share of flow	39.2%
Mean yard dwell	13.2 h
p95 yard dwell	49.0 h
Mean charging wait	60 min
Mean charger utilisation	14.8%
Vehicles not ready at dispatch	16,107
Late-dispatch ratio	42.9%
Clean-exit rate (on time and ready)	41.4%
Global readiness score	72.6
Principal bottleneck cause	Internal blocking and relocation
Area of greatest throughput loss	Logistics

B. EV versus ICE pressure scores

Driver	EV	ICE
Sequence disruption	40.3	31.5
Yard congestion	73.7	83.5
Charging pressure	16.0	0.0
Dispatch-delay risk	54.1	28.3
Launch-transition stress	44.1	34.4

C. Readiness by version

Propulsion	Version	Orders	Readiness
EV	VAN EV LARGE 01	5,594	28.6%
EV	VAN EV CREW 01	3,737	31.9%
EV	VAN EV MID 02	6,875	40.2%
EV	VAN EV MID 01	6,791	57.7%
ICE	VAN ICE LARGE 01	7,134	88.3%
ICE	VAN ICE CREW 01	5,429	89.6%
ICE	VAN ICE MID 02	10,713	92.6%
ICE	VAN ICE MID 01	12,424	96.9%

D. Capacity levers by expected impact

Lever	Expected impact
Charging capacity	0.37
EV sequencing	0.31
Yard management	0.29
Dispatch discipline	0.22
Shift resilience	0.20

E. Corrective package delta summary

Metric	Base	Corrective	Delta
Throughput	159.2	163.3	+2.6%
Internal time	1683.1	1592.3	-5.4%
Average yard occupancy	8.8%	8.3%	-5.8%
Peak yard occupancy	57.0%	55.0%	-3.5%
Charging wait (min)	67.1	69.8	+4.0%
Low-readiness dispatch risk	0.007	0.006	-4.3%
Congestion risk	0.097	0.091	-6.0%
Late vehicles	43.5%	40.2%	-7.5%
Operational stability	87.3	88.0	+0.8%

F. Ranking stability detail

Test	Result
Monte Carlo top-1: Logistics	77.3%
Monte Carlo top-1: Yard	22.7%
Sensitivity cases tested	10
Top-three pattern across sensitivity	Logistics, Yard, Charging; Logistics, Yard, Production

G. Figure index

Figure	Figure
Fig 1 Daily throughput	Fig 2 Weekly EV share
Fig 3 Dispatch funnel	Fig 4 EV vs ICE pressure
Fig 5 Readiness cohort	Fig 6 Delay concentration
Fig 7 Lead-time distribution	Fig 8 Readiness heatmap
Fig 9 Yard zone congestion	Fig 10 OPI ranking
Fig 11 Risk matrix	Fig 12 Driver correlation
Fig 13 Transition trend	Fig 14 Market geography
Fig 15 Scenario decision	Fig 16 Scenario trade-off
Fig 17 Before vs after	Fig 18 Lever ranking
Fig 19 Monte Carlo stability	